INCH-POUND
A-A-59006A
April 13, 2021
SUPERSEDING
A-A-59006
December 3, 1996

COMMERCIAL ITEM DESCRIPTION

INDICATORS, PIPELINE, FLOW, SIGHT

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers sight flow pipeline indicators intended for installation in water, fuel, and oil piping systems where visual indication is necessary or desired. These items are referred to as "sight flow indicators" herein.

2. CLASSIFICATION. Sight flow pipeline indicators shall conform to the following flange types, sizes, and pressure ratings.

2.1 Flange type.

a. ASME B16.5
b. MIL-PRF-20042

2.2 Size.

a. 1 inch
b. 2 inches
c. 3 inches
d. 4 inches
e. 6 inches
f. 8 inches

2.3 Pressure class.

a. 150
b. 250
c. 300

3. SALIENT CHARACTERISTICS.

3.1 Codes and standards. Sight flow indicators shall comply with the design, fabrication, testing, inspection, and certification requirements of ASME Boiler and Pressure Vessel Code, section VIII.

3.2 Design. Each sight flow indicator shall be as specified by table I. The sight flow indicator shall be designed so as not to obstruct flow of the fluid and shall not have an internal indicator.

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that improve this document should be sent to: Commander, Naval Sea Systems Command, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 4730

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TABLE I. Reference data.

Equipment/characteristic	Description
Fluid type	Seawater, fuel oil, lube oil
Operating pressure	400 pounds per square inch (psi) (max) (2758 kilopascal [kPa])
Hydrostatic test pressure	600 psi (max) (4137 kPa)
Temperature range	Normal 40 to 220 °F (4 to 104 °C)
Piping connections	Classes 150 and 300: ASME B16.5 raised-face flanges, straight through configuration
	Classes 150 and 250: MIL-PRF-20042 full-face flanges, straight through configuration
Shock resistance	MIL-DTL-901, grade A, class 1, type B
Vibration resistance	MIL-STD-167-1, type 1

3.3 Material. The material requirements for all sight flow indicator components shall be in accordance with table II. Sight flow indicators in seawater service shall utilize non-ferrous material.

TABLE II. Materials required for sight flow indicator.

Part	Material	Applicable documents	Remarks
Body	Stainless Steel	ASTM A351/A351M	Grade CF8M
	70-30 CuNi	ASTM B369/B369M	C96400
	Bronze	ASTM B61	C95800
		ASTM B62	
Glass	Borosilicate		Tempered
Threaded fasteners	CRES	NASM17829	
		ASTM F593	
	NiCu	ASTM F467	N04400 or N04405
		ASTM F468	
O-rings	Fluorocarbon	SAE AMS7276	Type 1, class 1

3.4 Construction. The sight flow indicator is composed of four basic components, including the body, the glass, the O-rings, and the cover. The body provides in-line attachment capability and rigidity to the sight flow indicator. The glass provides the viewing window to the process flow stream. O-rings are compressed to tightly seal the gap and prevent leaking between the glass and body. The cover provides a compression surface for the bolts to hold the gasket and glass tightly against the sight flow body.

3.5 Piping connections. Flanges shall be in accordance with the requirements of ASME B16.5 for stainless steel units and MIL-PRF-20042 for bronze units. Flanges shall be integrally cast with the body as applicable to the working pressure.

3.6 Envelope dimensions. Envelope dimensions shall be in accordance with table III and table IV. Metric conversions, when necessary for manufacturing processes, shall be in accordance with FED-STD-376 or ASTM SI10.

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TABLE III. Envelope dimensions for class 150 rated sight flow indicators.

Sight flow indicator size (inches)	Width (inches)	Flange-to-flange (inches)	View port inside diameter (I.D.) (inches)
1	5.750	5.625	1.500
2	9.313	7.875	2.875
3	11.938	9.375	3.750
4	11.625	11.000	4.750
6	18.000	14.250	7.000
8	21.000	16.125	9.000

TABLE IV. Envelope dimensions for classes 250 and 300 rated sight flow indicators.

Sight flow indicator size (inches)	Width (inches)	Flange-to-flange (inches)	View port I.D. (inches)
1	6.438	6.875	1.375
2	9.625	8.625	2.125
3	13.375	11.375	3.188
4	16.500	13.250	4.250
6	23.750	21.375	6.250
8	23.750	22.375	6.250

3.7 Identification plates. Identification plates shall be provided in accordance with the requirements of MIL-DTL-15024 and MIL-P-15024/5 and shall include the following:

a. CID number.
b. Manufacturer.
c. Contract number.
d. The nominal pipe size of the inlet and outlet connections.
e. Pressure rating.

3.7.1 Part numbering of interchangeable parts. The manufacturer's part number and drawing number shall be the same. All fasteners shall be identified by an industry part number which identifies both size and material.

3.8 Mechanical shock. The complete sight flow indicator assembly shall be capable of withstanding the medium impact shock test in accordance with 5.7.3.1.

3.9 Vibration. The complete sight flow indicator assembly shall be designed and tested to function in accordance with the requirements of MIL-STD-167-1, type 1 environmental vibration.

3.10 Workmanship. The body and flanges shall be of uniform quality and condition, as well as free from blow holes, porosity, hard spots, shrinkage defects, cracks, and other defects. All surfaces shall be smooth and well cleaned. The inside surfaces shall be free from sharp edges.

4. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 5.3).

b. Conformance inspection (see 5.4).

5.3 First article inspection. First article inspection shall consist of one sight flow indicator of each size and identical design undergoing the examinations and tests as specified in table V.

TABLE V. First article and conformance inspection.

Examinations and tests	First article	Conformance
Examination	5.6	5.6
Shock	5.7.3.1	---
Vibration	5.7.3.2	---
Hydrostatic	5.7.1	5.7.1
Burst test	5.7.2	---
Additional inspection	5.8	5.8

5.4 Conformance inspection. All assembled sight flow indicators shall undergo testing as specified in table V. Visual and dimensional examination shall consist of selecting sight flow indicators in accordance with 5.5 and inspecting them in accordance with 5.6.

5.5 Sampling for conformance inspection.

5.5.1 Lot. For sampling, a lot shall consist of sight flow indicators of the same size offered for delivery at one time.

5.5.2 Sampling. A random sample of flanges shall be selected from each lot offered to the government in accordance with ANSI/ASQ Z1.4 at inspection level II and the acceptance quality limit (AQL) of 1.0. Examination of 5.6 and the hydrostatic test of 5.7.1 shall be conducted upon these samples. If one or more defects are found in any sample, the entire lot shall be rejected. The contractor has the option of screening 100 percent of the rejected lot for the defective characteristic(s) or providing a new lot which shall be inspected in accordance with the sampling plan.

5.6 Visual and dimensional examination. The sight flow indicators shall be examined to verify conformance to the requirements of table VI. Any indicators having one or more defects shall be rejected.

TABLE VI. Visual and dimensional inspection.

Inspection	Paragraph
Construction	3.4
Piping connections	3.5
Identification plate	3.7
Quality of workmanship	3.10

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5.7 Tests.

5.7.1 Hydrostatic pressure test for strength and porosity. The test shall be an internal pressure test with water. Each assembled indicator shall be tested to a pressure equal to 150 percent of the design pressure for a duration of 60 minutes. Any indicator not maintaining zero leakage shall be rejected.

5.7.2 Burst test. The sight flow indicator shall be tested for compliance with requirement of table I in accordance with ASME Boiler and Pressure Vessel Code, section VIII. Any indicator that fails to meet the performance requirements of table I shall be rejected.

5.7.3 Mechanical tests.

5.7.3.1 Mechanical shock test. The sight flow indicator shall be tested for compliance with medium impact shock resistance in accordance with the requirements of MIL-DTL-901, grade A.

5.7.3.2 Vibration test. The sight flow indicator shall be tested for compliance with MIL-STD-167-1, type 1. Any unit not meeting MIL-STD-167-1, type 1 requirements shall be rejected.

5.8 Additional inspection. Unless otherwise specified in the contract or order (see 7.5), where other specifications form a part of this CID, sampling, examination, and tests shall be conducted as required by the pertinent specification.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PINs for these sight flow indicators are created as follows:

AA	**59006**	**-**	**X**	**-**	**X**	**-**	**XXX**	**X**
Prefix for CIDs	CID number	-	Flange type (see code below)	-	Size (see code below)	-	Pressure rating (150, 250, or 300)	Material (see code below)

Flange type	Code
ASME B16.5	N
MIL-PRF-20042	M

Size (inches)	Code
1	A
2	B
3	C
4	D
6	E
8	F

Material	Code
Stainless Steel	S
CuNi	C
Bronze	B

Example: AA59006-N-D-250S represents a 4-inch ASME flange, class 250 sight flow indicator made of stainless steel (see table II).

7.2 Source of documents.

7.2.1 ASME Standards. ASME standards are available from www.asme.org or American Society of Mechanical Engineers, 150 Clove Road, Little Falls, NJ 07424-2100.

7.2.2 ASTM Standards. ASTM standards are available from www.astm.org or ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959.

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7.2.3 Defense specifications and standards. Defense specifications and standards are available online at https://quicksearch.dla.mil.

7.2.4 NASM standards. NASM standards are available from www.aia-aerospace.org or Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3901.

7.2.5 SAE standards. SAE standards are available from www.sae.org or SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001.

7.3 First article. The first article consists of the units as specified in 5.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired and tested by the Government; and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

7.4 Preinstallation instructions. A set of instructions covering the preinstallation of the equipment should be furnished. Instructions should include all information necessary to return the unit to active status, such as, but not limited to the following: addition of lubricants prior to operation, flushing of lines, removal of greaseproof barrier, and the location of detached components. Instructions should be packaged in a transparent, waterproof, plastic bag that is a minimum 4 mils thick. Closure should be by heat sealing. The shipping container in which the instructions are packed should be so marked.

7.5 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Quantity required.
c. Product conformance provisions (see 5.1).
d. Packaging requirements (see 6).
e. First article requirements, if applicable (see 7.3).
f. Preinstallation instructions required (see 7.4).

7.6 Key words.

Borosilicate

Strainer

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – CR4
Navy – SH
Air Force – 71

Review Activities:
Army – GL
Navy – AS, OS, YD
GSA – FAS

Preparing Activity:
Navy – SH
(Project 4730-2021-037)

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